POLYP DETECTION MEDICAL REPORT

Analysis Date: June 03, 2026

VIDEO INFORMATION

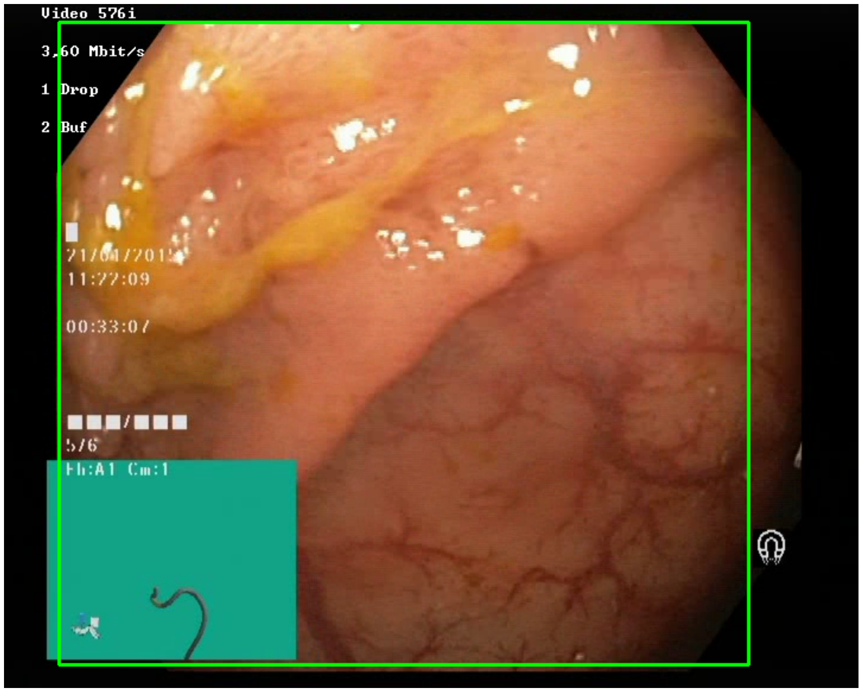
Video ID:	1de3ef0f-d54d-49e3-98a1-a919c5552a64
Total Frames:	1216
Frame Rate:	25.0 fps (assumed)
Duration:	48.6 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	206	✓
RT-DETR Detections	295	✓
MedSAM2 Detections	303	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 164 frames)



Feature	Value	Interpretation
Frame Index	978	Frame 978 of 1216
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**MALIGNANT_POLYP**	Type confidence: 92.0%
Redness Score	0.193	Color-based vascularization
Relative Radius	66.7%	Polyp size as % of frame diagonal
Texture Score	0.880	Surface morphology
Vessel Visibility	0.157	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output

Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	76.7%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 33 frames)

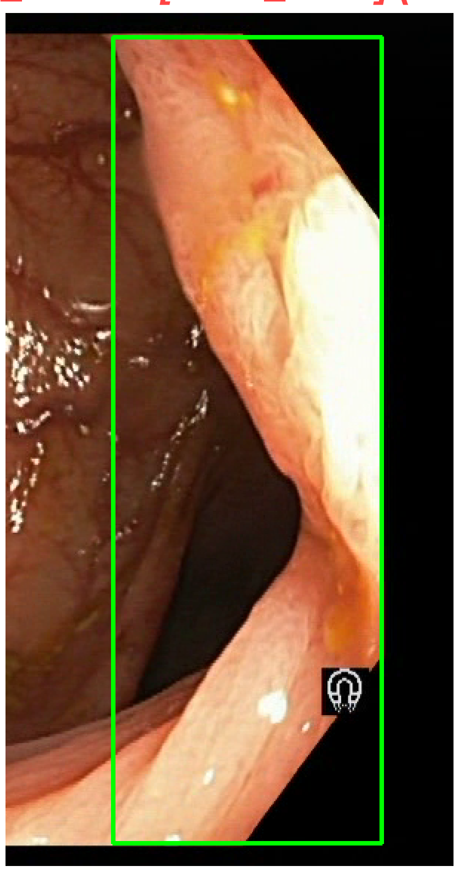
Feature	Value	Interpretation
Frame Index	1205	Frame 1205 of 1216
Detection Method	Detector Consensus (YOLO + RT-DETR agreement)	(MedSAM2 unavailable for this video)
POLYP TYPE	**MALIGNANT_POLYP**	Type confidence: 92.0%
Redness Score	0.180	Color-based vascularization

Relative Radius	36.9%	Polyp size as % of frame diagonal
Texture Score	1.000	Surface morphology
Vessel Visibility	0.332	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	73.3%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #3 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 18 frames)



Feature	Value	Interpretation

Frame Index	1129	Frame 1129 of 1216
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.236	Color-based vascularization
Relative Radius	27.1%	Polyp size as % of frame diagonal
Texture Score	0.089	Surface morphology
Vessel Visibility	0.931	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	70.4%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:
A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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